

Introduction to Genetics

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Definitions

Genetics (The term Genetic derives from ancient Greek *Genetikos* and that is from genesis which means” origin”)
-is a branch of biology that deals with the study of molecular structure and function of genes, patterns of inheritance from parent to offspring(heredity), gene distribution and variation in populations.

Medical Genetics:

Is a branch of medicine dealing with the inheritance of diseases in families (using **pedigree analysis**), diagnosis and treatment of diseases due to single gene mutation, chromosome abnormalities, and multifactorial disorders.

Medical genetics involves any application of genetics for medical practices(mapping disease causing genes on specific chromosome, genetic counseling and genetic testing).

The importance of Genetics:

-Genetics has close connections with our everyday modern life, it contributes to **medicine, agriculture, industry , technology, law, military, politics, forensics and ethics.**

- Every person is unique. Part of what makes you unique is your genes. We all possess genes that influence our lives. They affect our **height, weight, hair color, and skin pigmentation**. Genes can increase our susceptibility to many diseases and disorders and even contribute to our **intelligence, talents and personality**.

- Today, the major **crop plants** and **animals** used in agriculture have undergone extensive genetic alterations or modifications to greatly increase their yields(quantitatively) and provide many desirable traits(qualitatively), such as plants **resistance to drought, viral diseases, herbicides and pests**, in addition to special nutritional qualities, substances of commercial value such as a genetically modified strain of rice (The **golden rice** manufactures as much vitamin A precursor and iron as unrelated rice).

-Growth hormone, insulin, safer vaccine, clotting factor, enzymes ,erythropoietin, interferons are now produced commercially by genetically engineered bacteria.

-Genetics also plays a critical role in medicine.

Physicians recognize that many diseases and disorders have hereditary background, including well-known disorders such as sickle-cell anemia, hemophilia, Thalassemia and Huntington disease as well as many common diseases such as asthma, diabetes, and hypertension.

-Advances in **molecular genetics** have allowed the development of many diagnostic tests. **Gene therapy** (the replacement of a missing gene product by inserting a normal gene into a somatic cell of patients in order to correct genetic diseases) has now been carried out on thousands of patients.

Terminology:

Gene: The basic unit of heredity in the chromosomes; a sequence of nucleotides in a DNA molecule that performs a specific function, such as coding for RNA molecules or polypeptides.

Genome: The complete set of chromosomes with their associated genes. .(all genetic instructions in a haploid cell)

Alleles: Two or more different forms of a gene. Alleles occupy the same position (locus) on **homologues chromosomes** and are separate from each other at meiosis.

Genotype: the sum total of all the genes present in an individual.

Phenotype: Observable characteristics of an organism, resulting from interactions between the **genotype** and **environment**.

Heredity: The transmission of characteristics from parents to offspring.

Karyotype: The general appearance of metaphasic chromosomes of an organism with regard to number, size, shape, centromere's locations, and banding patterns.

Mutation: A permanent change in a gene from one allelic form to another; an inheritable change in nucleotide sequence of chromosomes.

Mutant: An organism carrying a gene that has undergone a mutation.

Mutagen: A chemical, physical or biological agent that increases the mutation rate.

Teratogen: A teratogen is an agent that can produce a permanent alteration in structure or function of an organism after exposure during embryonic or fetal life. Teratogens include environmental factors ,medication ,drug of abuse, and chemicals.

Recessive allele: An allele whose phenotypic effect is masked in the heterozygous by that of another, dominant allele.

Dominant allele: An allele whose phenotypic effect is the same in both heterozygous and homozygous conditions.

Haploid: The number of chromosomes in the gametes is referred to as **haploid** (single set), and the number in the somatic cells as the **diploid** (double sets). Cells that have more than two sets of chromosomes are known as **polyploidy** (many sets).

Penetrance: Is the proportion of individuals having a particular genetic variant (such as mutation in a specific gene) that express the expected phenotype. Some times despite the presence of a dominant allele, a phenotype may not be present(incomplete penetrance).e.g. Polydactyly or extra fingers and toes .

Expressivity: Is the degree to which a character is expressed (variations in phenotypic expression or range of signs and symptoms).